AWS Migration Business Case

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Executive Summary

This AWS migration assessment analyzed 9 virtual machines with 25 vCPUs and 112 GiB of memory across Windows, RHEL, and Linux environments. The migration to AWS presents **estimated annual savings of up to $4,230 (30%) compared to on-demand pricing** through strategic use of Reserved Instances, while optimizing resource allocation from 25 vCPUs to 20 vCPUs and reducing memory footprint from 112 GiB to 54 GiB.

Assessment Scope:

- 9 virtual machines analyzed for AWS migration
- Mixed OS environment: 4 Windows, 2 RHEL, 3 Linux servers
- 875 GiB of storage mapped to AWS EBS volumes
- Right-sized to 6 distinct EC2 instance types optimized for workload requirements

Strategic Benefits:

Benefit	Impact	Value
Cost Optimization	Annual savings through Reserved Instances	Up to $4,230 (30%)
Resource Efficiency	Right-sized compute allocation	20% vCPU reduction, 52% memory optimization
Operational Excellence	Managed infrastructure with AWS services	Reduced maintenance overhead
Scalability	On-demand resource scaling capabilities	Elastic capacity management
Reliability	AWS infrastructure with 99.99% SLA	Enhanced uptime and disaster recovery

The analysis demonstrates that a 3-year Reserved Instance commitment delivers the highest ROI with estimated total annual costs of $10,121 compared to $14,351 on-demand pricing.

Insights

Summary

Cost Component	On-Demand	1-Year NURI (Shared Tenancy)	3-Year NURI (Shared Tenancy)
Compute	$13,511	$10,789	$9,281
Storage	$840	$840	$840
Annualized Total	$14,351	$11,629	$10,121
Annual Savings against On-Demand	-	19%	30%

The cost analysis reveals significant savings opportunities through Reserved Instance commitments. The 3-year NURI option provides the optimal balance of cost savings and commitment, delivering 30% annual savings compared to on-demand pricing. Storage costs remain consistent across all pricing models at $840 annually for 875 GiB of EBS volumes.

Instance Type Distribution:

- **2 c5a.xlarge instances** (compute-optimized): Highest cost component at $5,922 on-demand, ideal for CPU-intensive workloads
- **3 c7a.medium instances** (latest generation compute): Cost-effective for smaller compute workloads at $1,349 on-demand
- **1 m5a.xlarge instance** (general purpose): Balanced compute and memory at $3,119 on-demand
- **1 m5a.large instance** (general purpose): Mid-tier workload support at $1,559 on-demand
- **1 m7a.medium instance** (latest generation general purpose): Optimized performance at $634 on-demand
- **1 c5a.large instance** (compute-optimized): Specialized workload support at $929 on-demand

The recommended instance mix provides optimal price-performance ratios while maintaining compatibility with existing workload requirements across Windows, RHEL, and Linux operating systems.

Next Steps

1. **Engage Specialists**: [Engage a specialist] if you want further support and guidance on your migration journey.

2. **Perform wave planning and migration**: Utilize the AWS Transform VMware job capabilities to perform comprehensive application assessment through dependency mapping, create migration waves, and execute the VMware-to-EC2 migrations

Disclaimer

This report provides an estimate of fees in USD, and savings based on certain information you provide. Fee estimates do not include any taxes that might apply. Your actual fees and savings depend on a variety of factors, including your actual usage of AWS services, which may vary from the estimates provided in this report. Additional configurations are available on request by engaging a specialist.

Appendix

Appendix A: Assumptions

- Compute
  - AWS Region Used for Cost Modelling: us-east-1
  - Peak CPU Utilization (if missing data): 25%
  - Peak Memory Utilization (if missing data): 60%
  - Storage Utilization (if missing data): 50%
  - Provisioned Storage (if missing data): 500 GiB
  - SQL Server and Windows Server BYOL eligibility subject to Microsoft licensing terms
  - Windows Server BYOL only available on dedicated hosts for versions prior to Windows Server 2022

Appendix B: Distribution of Instances Recommended by Family

Shared Tenancy Instance Family Distribution

Instance Family	Number of Instances	Annualized 1yr NURI Costs
c5a	3	$5,598
m5a	2	$3,837
c7a	3	$893
m7a	1	$462